Nsight Quick Start

Installing nsight

```
# Download
cd ~
mkdir -p ~/installers && cd ~/installers

wget -O NsightSystems.run \
https://developer.nvidia.com/downloads/assets/tools/secure/nsight-
systems/2026_1/NsightSystems-linux-public-2026.1.1.204-3717666.run

# Make executable
chmod +x NsightSystems.run

# Extract to user directory
mkdir -p ~/tools/nsight-systems
./NsightSystems.run --target ~/tools/nsight-systems --noexec

NSYS_DIR=$(dirname $(find ~/tools/nsight-systems -type f -name nsys -perm -111 | head -n1))
echo "export PATH=$NSYS_DIR:\$PATH" >> ~/.bashrc
source ~/.bashrc

# Verify
nsys --version
```

Profiling

Follow the specification in the assignment, and implement code to isolated kernels using nvtx (see pg.5)

Example:

```
uv run nsys profile --trace nvtx \
                    --stats=true \
                    --output=nsight_log/profile_both \
                    --force-overwrite=true \
                    python student/benchmark.py \
                    --profile both \
                    --seq_length 128 \
                    --n_steps 100
```

- Add `--stats=true` and `--trace nvtx` to record the nvtx customized range (attention, matmul, softmax) so that you can view in the terminal output or through nsight desktop application. (You can also add `cuda` to `--trace` to track more terms)
- Add `--force-overwrite=true` if you save the output to the same path to overwrite (or you can view in

terminal or save as text)

- Parameters after python .. are customized (depend on your implementation).

View stats on the NVIDIA nsight desktop system (if you want to)

1. Download application from https://developer.nvidia.com/nsight-systems/get-started to your computer
2. Download the `.nsys-rep` output file to your local machine from hpc
3. Open the file in the nsight desktop, and you can view the time distribution and the statistics (under "Stats Systems View").

NVTX GPU Projection Summary
NVTX GPU Projection Trace
NVTX Push/Pop Range Summary
NVTX Push/Pop Range Trace
NVTX Range Kernel Summary
NVTX Range Summary
NVTX Start/End Range Summary
Network Devices Congestion
NvVideo API Summary
OS Runtime Summary
OpenACC Summary
OpenGL KHR_debug GPU Range Summar
OpenGL KHR_debug Range Summary
OpenMP Summary

Time	Total Time	Instances	Avg	Med	Min	Max	StdDev	Range
37.9%	5.653 s	100	56.535 ms	51.262 ms	50.682 ms	547.354 ms	49.593 ms	:backward_iteration
20.5%	3.064 s	100	30.644 ms	17.422 ms	17.289 ms	1.335 s	131.711 ms	:forward_iteration
16.5%	2.456 s	100	24.565 ms	20.390 ms	20.141 ms	426.165 ms	40.569 ms	:backward_pass_in_backward_iteration
12.7%	1.897 s	100	18.967 ms	18.785 ms	18.488 ms	27.657 ms	959.287 µs	:forward_pass_in_backward_iteration
7.5%	1.125 s	100	11.253 ms	11.151 ms	11.013 ms	16.565 ms	604.727 µs	:optimizer_step
2.3%	345.310 ms	1200	287.758 µs	196.477 µs	169.268 µs	110.572 ms	3.186 ms	:scaled_dot_product_attention
1.5%	216.605 ms	1200	180.504 µs	107.505 µs	91.922 µs	87.822 ms	2.532 ms	:softmax
1.0%	154.856 ms	1200	129.046 µs	129.494 µs	105.975 µs	2.696 ms	75.703 µs	:einsum